Operating Systems

UE23CS242B

4th Semester, Academic Year 2023

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WRITE A C PROGRAM TO LIST ALL FILES WHOSE NAME STARTS WITH A DIGIT (0-9). INPUT: DIRECTORY NAME AS A RUNTIME ARGUMENT.

CODE:

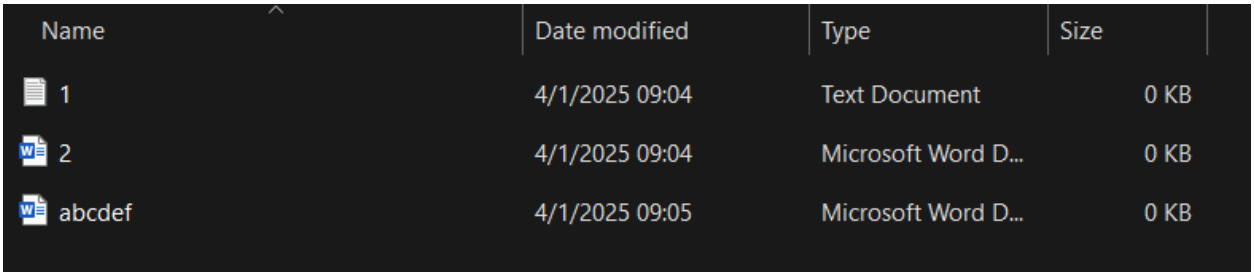
```c
// Write a C program to list all files whose name starts with a digit (0-9). Input: directory name as a runtime argument.
#include <stdio.h>
#include <stdlib.h>
#include <dirent.h>
#include <ctype.h>
#include <string.h>

int main(int argc, char *argv[]) {
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <directory_name>\n", argv[0]);
        return 1;
    }

    struct dirent *entry;
    DIR *dir;
    int count = 0;
    dir = opendir(argv[1]);
    if (dir == NULL) {
        perror("opendir");
        return 1;
    }
    while ((entry = readdir(dir)) != NULL) {
        if (isdigit(entry->d_name[0])) {
            count++;
            if (count == 1) {
                printf("All files whose name starts with a digit (0-9) in the directory %s are:\n", argv[1]);
            }
            printf("%s\n", entry->d_name);
        }
    }
    if (count == 0) {
        printf("There are no files whose name starts with a digit (0-9) in the directory %s.\n", argv[1]);
    }
    closedir(dir);
    return 0;
}
```

OUTPUT:

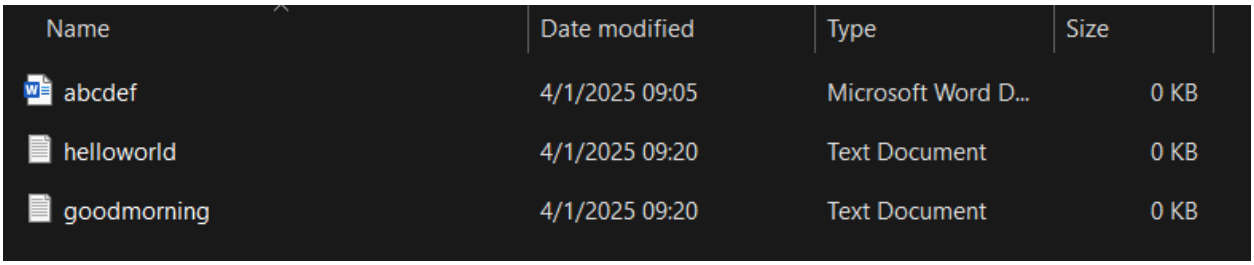
Folder1



Name	Date modified	Type	Size
1	4/1/2025 09:04	Text Document	0 KB
2	4/1/2025 09:04	Microsoft Word D...	0 KB
abcdef	4/1/2025 09:05	Microsoft Word D...	0 KB

```
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/Documents/pes/Sem 4/OS/codes# gcc assignment4.c -o assignment4
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/Documents/pes/Sem 4/OS/codes# ./assignment4 folder1
All files whose name starts with a digit (0-9) in the directory folder1 are:
1.txt
2.docx
```

Folder2



Name	Date modified	Type	Size
abcdef	4/1/2025 09:05	Microsoft Word D...	0 KB
helloworld	4/1/2025 09:20	Text Document	0 KB
goodmorning	4/1/2025 09:20	Text Document	0 KB

```
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/Documents/pes/Sem 4/OS/codes# gcc assignment4.c -o assignment4
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/Documents/pes/Sem 4/OS/codes# ./assignment4 folder2
There are no files whose name starts with a digit (0-9) in the directory folder2.
```